

Java Assignments

1. Write a Java program to get the character at the given index within the String.

Sample Output:

```
Original String = Java Exercises!  
The character at position 0 is J  
The character at position 10 is i  
public class CharAt {  
    public static void main(String[] args) {  
        String str = "Java Exercises!";  
  
        // Retrieve characters at position 0 and 10  
        char index0 = str.charAt(0);  
        char index10 = str.charAt(10);  
  
        // Display output  
        System.out.println("Original String = " + str);  
        System.out.println("The character at position 0 is " + index0);  
        System.out.println("The character at position 10 is " + index10);  
    }  
}
```

2. Write a Java program to compare two strings lexicographically. Two strings are lexicographically equal if they are the same length and contain the same number of characters in the same positions.

Sample Output:

```
String 1: This is Exercise 1  
String 2: This is Exercise 2  
"This is Exercise 1" is less than "This is Exercise 2"  
  
public class StringCompare {  
    public static void main(String[] args) {  
        String str1 = "This is Exercise 1";  
        String str2 = "This is Exercise 2";  
  
        System.out.println("String 1: " + str1);  
        System.out.println("String 2: " + str2);  
  
        // compareTo returns a value less than 0 if str1 is  
        lexicographically less than str2  
        int result = str1.compareTo(str2);  
  
        if (result < 0) {  
            System.out.println "\"" + str1 + "\" is less than \"" + str2 +  
            "\"");  
        } else if (result == 0) {  
            System.out.println "\"" + str1 + "\" is equal to \"" + str2 +  
            "\"");  
        } else {  

```

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```
        System.out.println "\"" + str1 + "\" is greater than \"" + str2
+ "\"");
    }
}
}
```

3. Write a Java program to check whether a given string ends with the contents of another string.

Sample Output:

```
"Python Exercises" ends with "se"? false
"Python Exercise" ends with "se"? true
public class EndsWith {
    public static void main(String[] args) {
        String str1 = "Python Exercises";
        String str2 = "Python Exercise";
        String endStr = "se";

        boolean result1 = str1.endsWith(endStr);
        boolean result2 = str2.endsWith(endStr);

        System.out.println "\"" + str1 + "\" ends with \"" + endStr + "\"? " +
result1);
        System.out.println "\"" + str2 + "\" ends with \"" + endStr + "\"? " +
result2);
    }
}
```

4. Write a Java program to get the index of all the characters of the alphabet.

Sample Output:

```
a  b  c  d  e  f  g  h  i  j
=====
36 10 7 40 2 16 42 1 6 20

k  l  m  n  o  p  q  r  s  t
=====
8 35 22 14 12 23 4 11 24 31

u  v  w  x  y  z
=====
5 27 13 18 38 37
```

Sample string of all alphabet: "The quick brown fox jumps over the lazy dog."

```
public class AlphabetIndex{
    public static void main(String[] args) {
        String str = "The quick brown fox jumps over the lazy dog.";
```

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```
for (char ch = 'a'; ch <= 'z'; ch++) {
    System.out.print(ch + " ");
}
System.out.println("\n=====");

for (char ch = 'a'; ch <= 'z'; ch++) {
    int index = str.indexOf(ch);
    System.out.print(index + " ");
}
System.out.println("\n\nSample string of all alphabet: \"" + str +
"\");
    }
}
```

5. Write a Java program to replace each substring of a given string that matches the given regular expression with the given replacement.

Sample string : "The quick brown fox jumps over the lazy dog."

In the above string replace all the fox with cat.

Sample Output:

Original string: The quick brown fox jumps over the lazy dog.
New String: The quick brown cat jumps over the lazy dog.

```
public class ReplaceString {
    public static void main(String[] args) {
        String str = "The quick brown fox jumps over the lazy dog.";
        String newStr = str.replaceAll("fox", "cat");

        System.out.println("Original string: " + str);
        System.out.println("New String: " + newStr);
    }
}
```

6. Write a Java program to convert all the characters in a string to uppercase.

Sample Output:

Original String: The Quick BroWn FoX!
String in uppercase: THE QUICK BROWN FOX!

```
public class UpperCase{
    public static void main(String[] args) {
        String str = "The Quick Brown Fox!";
        String upperStr = str.toUpperCase();

        System.out.println("Original String: " + str);
        System.out.println("String in uppercase: " + upperStr);
    }
}
```

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7. Write a Java program to reverse a string.

Sample Output:

```
The given string is: The quick brown fox jumps  
The string in reverse order is:  
spmuj xof nworb kciuq ehT
```

```
public class Reverse {  
    public static void main(String[] args) {  
        String str = "The Quick Brown Fox!";  
  
        StringBuilder sb = new StringBuilder(str);  
        String reversedStr = sb.reverse().toString();  
  
        System.out.println("Original String: " + str);  
        System.out.println("Reversed String: " + reversedStr);  
    }  
}
```